ALLIANCE UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

DATABASE MANAGEMENT SYSTEM

ASSIGNMENT 6

IMPLEMENTATION OF DATA INTEGRITY CONSTRAINTS

Hotel Management System

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Programme	BCA – Artificial Intelligence & Machine Learning
Subject	Database Management System
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Academic Year	2nd Year

Introduction

Integrity constraints are rules that help maintain the accuracy, validity, and consistency of database records. This assignment applies primary key, foreign key, UNIQUE, CHECK, and DEFAULT constraints to the Hotel Management System using MySQL 8.0.

1. Primary Key and Foreign Key Constraints

Table	Primary Key	Foreign Key
Guest	guest_id	—
Room_Type	room_type_id	—
Room	room_id	room_type_id → Room_Type.room_type_id
Staff	staff_id	—
Reservation	reservation_id	guest_id → Guest.guest_id; room_id → Room.room_id
Service	service_id	—
Payment	payment_id	reservation_id → Reservation.reservation_id
Service_Usage	usage_id	reservation_id → Reservation.reservation_id; service_id → Service.service_id

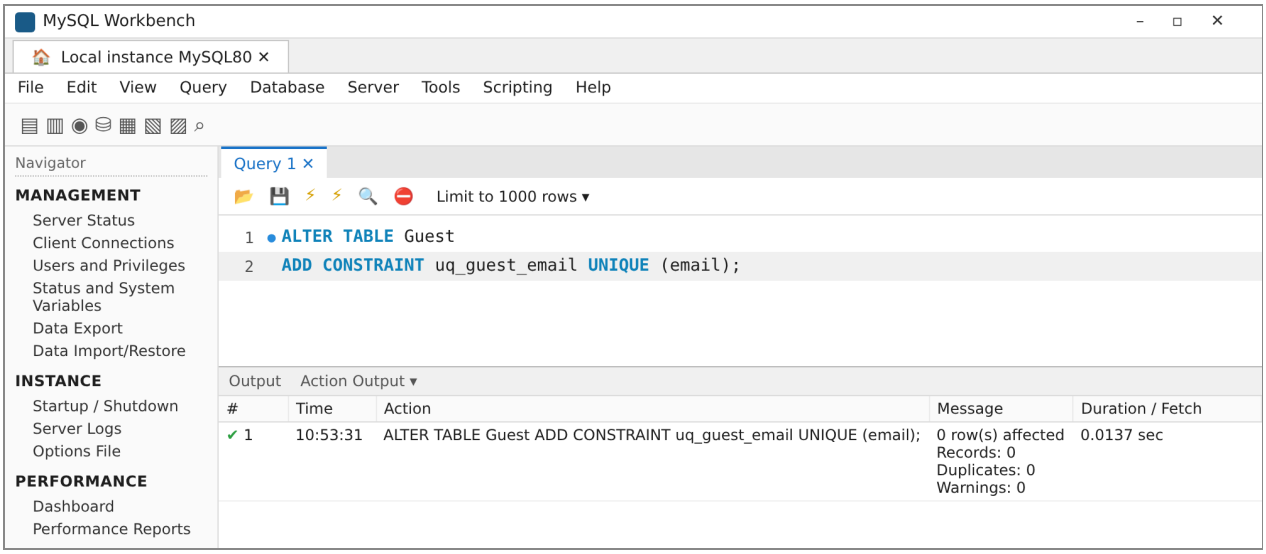
A primary key uniquely identifies a row. A foreign key ensures that a reference points to an existing record in the related table.

2. Implementing UNIQUE, CHECK, and DEFAULT Constraints

A. UNIQUE Constraint

A UNIQUE constraint prevents duplicate values in a column. If each guest should have a distinct email address, the following constraint can be used:

```
ALTER TABLE Guest
ADD CONSTRAINT uq_guest_email UNIQUE (email);
```



This assumes that guest email addresses are intended to be unique. MySQL permits multiple NULL values in a UNIQUE column.

B. CHECK Constraints

CHECK constraints restrict the values that can be stored in a column.

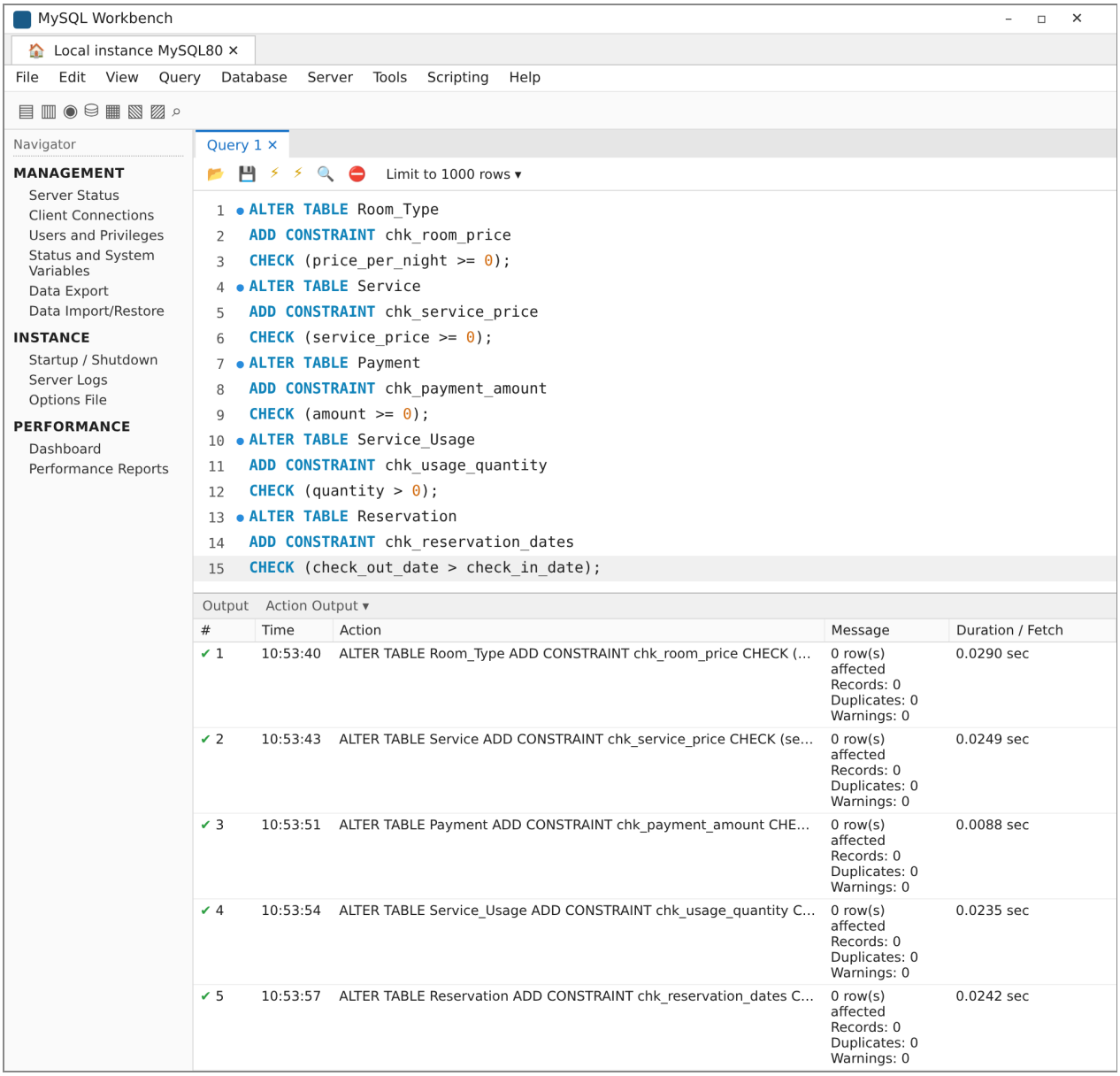
```
ALTER TABLE Room_Type
ADD CONSTRAINT chk_room_price
CHECK (price_per_night >= 0);

ALTER TABLE Service
ADD CONSTRAINT chk_service_price
CHECK (service_price >= 0);

ALTER TABLE Payment
ADD CONSTRAINT chk_payment_amount
CHECK (amount >= 0);

ALTER TABLE Service_Usage
ADD CONSTRAINT chk_usage_quantity
CHECK (quantity > 0);

ALTER TABLE Reservation
ADD CONSTRAINT chk_reservation_dates
CHECK (check_out_date > check_in_date);
```



- Room-type prices and service prices cannot be negative.
- Payment amounts cannot be negative.
- Service usage quantity must be greater than zero.
- A reservation's check-out date must be later than its check-in date.

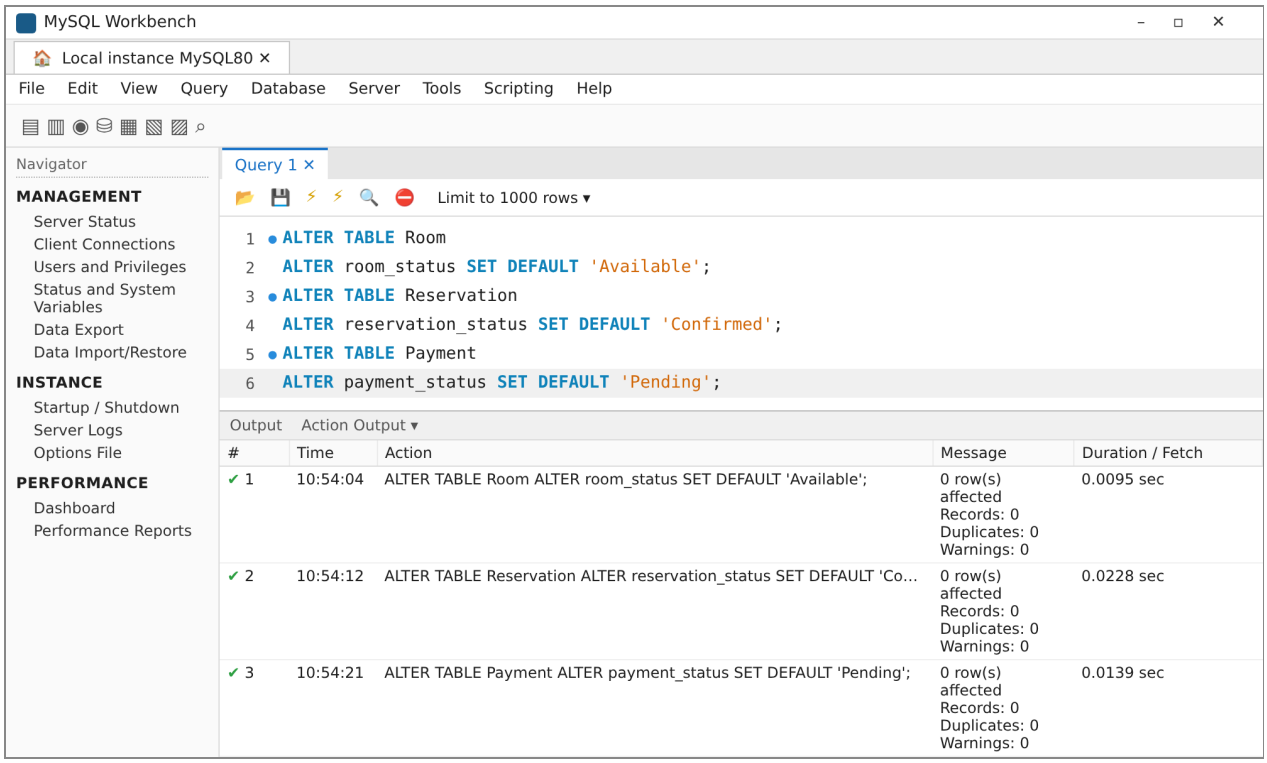
C. DEFAULT Constraints

A DEFAULT constraint supplies a value when an INSERT statement omits that column.

ALTER TABLE Room
ALTER room_status SET DEFAULT 'Available';

ALTER TABLE Reservation
ALTER reservation_status SET DEFAULT 'Confirmed';

ALTER TABLE Payment
ALTER payment_status SET DEFAULT 'Pending';



These are MySQL 8.0 column-alteration forms. Run each statement only once; if the same constraint or default already exists, inspect the table definition before applying it again.

3. Testing the Constraints

SHOW CREATE TABLE Guest;
SHOW CREATE TABLE Room_Type;
SHOW CREATE TABLE Room;
SHOW CREATE TABLE Reservation;
SHOW CREATE TABLE Service;
SHOW CREATE TABLE Payment;
SHOW CREATE TABLE Service_Usage;

MySQL Workbench

Local instance MySQL80

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Query 1

Limit to 1000 rows

```
1  SHOW CREATE TABLE Guest;
```

Table	Create Table
guest	CREATE TABLE `guest` (`guest_id` int NOT NULL, `guest_name` varchar(100) NOT NULL, `email` varchar(100) DEFAULT NULL, `phone` varchar(15) DEFAULT NULL, `address` varchar(200) DEFAULT NULL, PRIMARY KEY (`guest_id`), UNIQUE KEY `uq_guest_email` (`email`)) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci

Result 1 Read Only

Output Action Output

#	Time	Action	Message	Duration / Fetch
✔ 1	10:54:28	SHOW CREATE TABLE Guest;	1 row(s) returned	0.0389 sec / 0.000056 sec

MySQL Workbench

Local instance MySQL80

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Query 1

Limit to 1000 rows

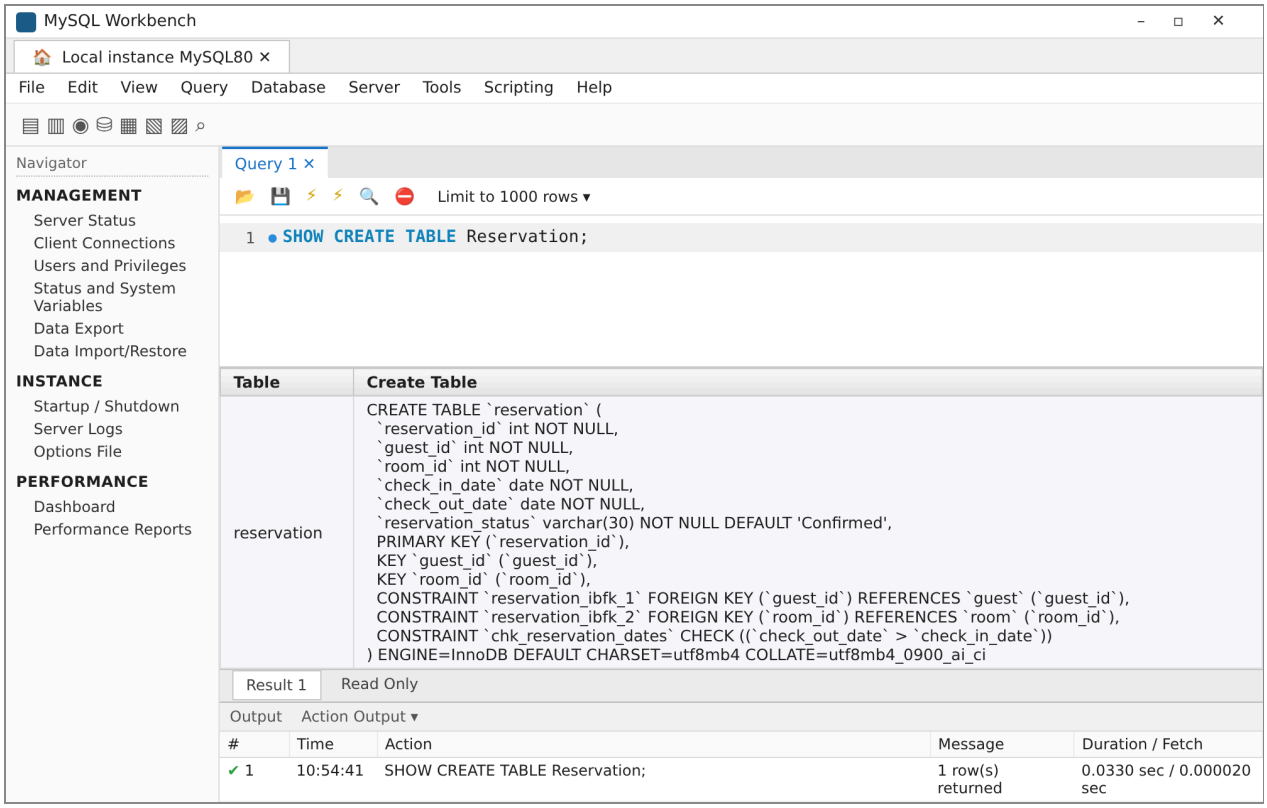
```
1  SHOW CREATE TABLE Room_Type;
```

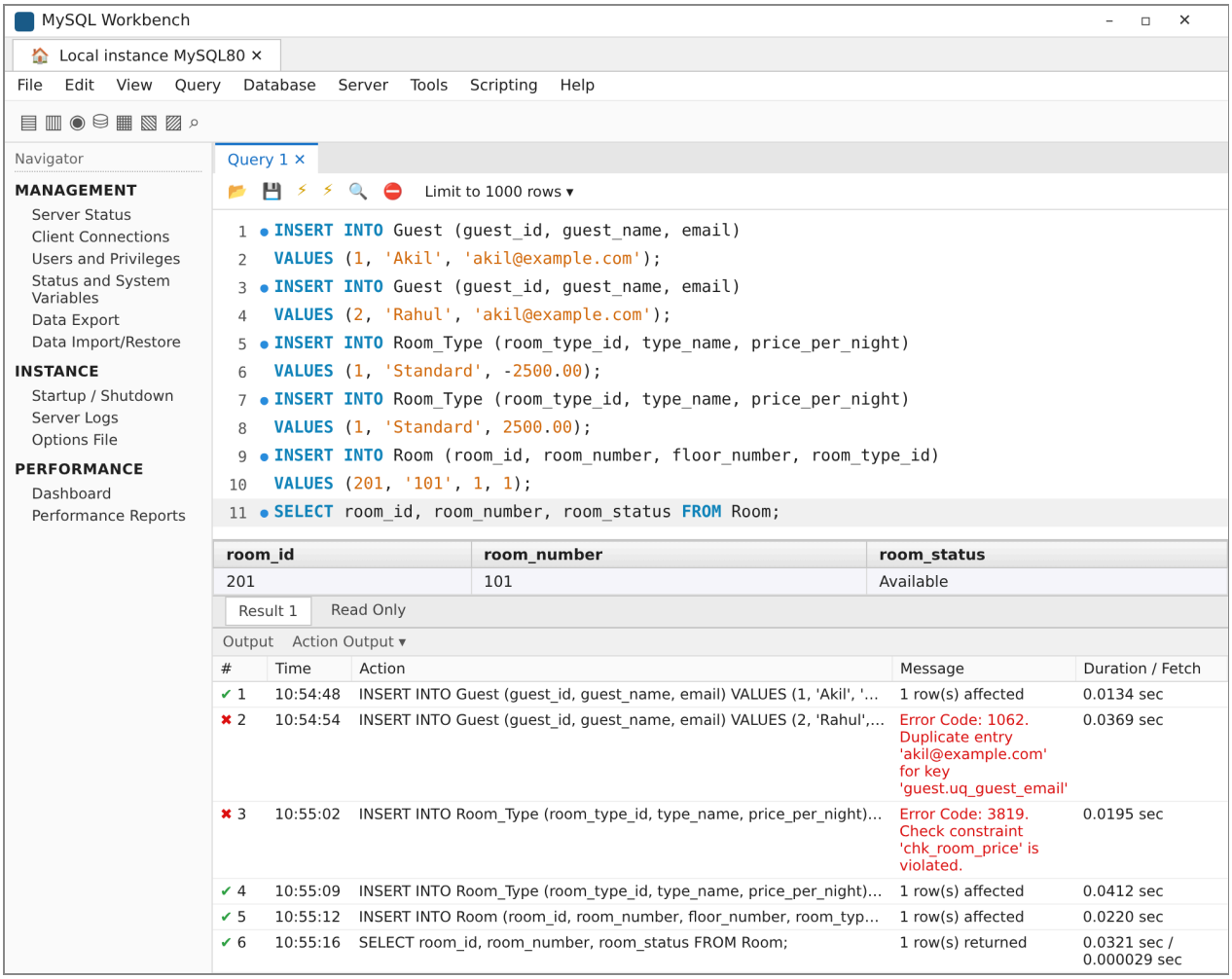
Table	Create Table
room_type	CREATE TABLE `room_type` (`room_type_id` int NOT NULL, `type_name` varchar(50) NOT NULL, `price_per_night` decimal(10,2) NOT NULL, `description` varchar(200) DEFAULT NULL, PRIMARY KEY (`room_type_id`), CONSTRAINT `chk_room_price` CHECK ((`price_per_night` >= 0))) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci

Result 1 Read Only

Output Action Output

#	Time	Action	Message	Duration / Fetch
✔ 1	10:54:33	SHOW CREATE TABLE Room_Type;	1 row(s) returned	0.0113 sec / 0.000033 sec


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Query 1
Limit to 1000 rows
1 SHOW CREATE TABLE Reservation;
Table
Create Table
reservation
CREATE TABLE `reservation` (
`reservation_id` int NOT NULL,
`guest_id` int NOT NULL,
`room_id` int NOT NULL,
`check_in_date` date NOT NULL,
`check_out_date` date NOT NULL,
`reservation_status` varchar(30) NOT NULL DEFAULT 'Confirmed',
PRIMARY KEY (`reservation_id`),
KEY `guest_id` (`guest_id`),
KEY `room_id` (`room_id`),
CONSTRAINT `reservation_ibfk_1` FOREIGN KEY (`guest_id`) REFERENCES `guest` (`guest_id`),
CONSTRAINT `reservation_ibfk_2` FOREIGN KEY (`room_id`) REFERENCES `room` (`room_id`),
CONSTRAINT `chk_reservation_dates` CHECK ((`check_out_date` > `check_in_date`))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
Result 1
Read Only
Output
Action Output
Time Action Message Duration / Fetch
1 10:54:41 SHOW CREATE TABLE Reservation; 1 row(s) returned 0.0330 sec / 0.000020 sec



The output displays each table definition, including applicable keys and constraints.

Expected Result

- Duplicate non-NULL guest email addresses are rejected if uq_guest_email is added.
- Negative prices and payment amounts are rejected.
- Zero or negative service quantities are rejected.
- A check-out date equal to or earlier than the check-in date is rejected.
- If a status column is omitted during insertion, the configured default is used.

Conclusion

Integrity constraints make the Hotel Management System database more reliable. Primary keys uniquely identify records, foreign keys maintain relationships, UNIQUE constraints prevent duplicate values, CHECK constraints restrict invalid data, and

DEFAULT constraints provide initial values. Together, these rules improve consistency and help prevent incorrect records.